YIMIN TANG

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Github: https://github.com/TachikakaMin, Website: https://sites.google.com/view/yimintang

EDUCATION

University of Southern California
PhD in Computer Science, Advisor: Sven Koenig, Erdem Bıyık
Viterbi School of Engineering

Carnegie Mellon University
Master of Science in Robotics, Advisor: Katia Sycara, Jiaoyang Li
Robotics Institute, School of Computer Science (RI, SCS)

ShanghaiTech University
B.E in Computer Science and Technology
The School of Information Science and Technology (SIST)

RESEARCH KEYWORDS

Robotics, Path/Motion Planning, Auto Research, LLM, Deep RL, Imitation Learning, Python, C++

WORK EXPERIENCE

Symbotic, Routing *May 2026 - August 2026*
Applied Scientist, Auto Research, mentored by Jingkai Chen

- Developed a novel algorithm that unifies task and path planning, resulting in a peer-reviewed publication and a patent filing.
- Adapted and validated the algorithm on Symbotic's physical warehouse robots, bridging simulation and real-world deployment.
- Built an LLM-driven autonomous research and optimization pipeline that improved end-to-end planning throughput by 150%.

Stripe, Machine Learning *May 2025 - August 2025*
Machine Learning Researcher, Agent, mentored by Yiing Chau Mak

- Supported a customer-support LLM system using retrieval-augmented generation (RAG) grounded in Stripe documentation.
- Designed an iterative feedback loop to continuously improve retrieval accuracy and response quality.
- Built a QA benchmark to evaluate the system, demonstrating a 5% reduction in customer dissatisfaction and improved retrieval of customer-specific information.

Futurewei Technologies, Machine Learning *May 2024 - May 2025*
LLM Researcher, Agent, mentored by Dr.Masood Mortazavi and Dr.Ning Yan

- Designed Turing machine-inspired algorithms for long-context LLMs, integrating RAG with novel Memory mechanisms.
- Achieved a 100% success rate on the Multi-Needle-in-a-Haystack (M-NIAH) benchmark with a 500K-token context window using LLaMA 3 70B.
- Published the resulting research at NeurIPS 2024 Workshop.

Microsoft, AzureStack Storage *August 2020 - August 2021*
Software Engineer, Full Time Engineer

- Designed and validated new features for Azure Blob, Table, and Queue Storage modules using C# and C++.
- Led the Azure Stack Storage API upgrade, coordinating implementation, compatibility validation, and integration.
- Investigated intermittent, hard-to-reproduce bugs with incomplete logs through systematic root-cause analysis.

OPEN SOURCE (PR MERGED)

PyTorch, Taichi, dreamer-torch, TrackFormer, Azure Java SDK, OI Wiki, onscripter-GBK-2015, MuJoCo, PlanViz, FFXIVChnTextPatch-Souma

SERVICE

Reviewer for T-RO, ICRA, IROS, CVPR, ICLR, ICML, NeurIPS, CoRL, AAAI, WAFR, ICAPS, ECAI, AISTATS, JAIR, IEEE SMC, RA-L, TJSC
Organizer for The League of Robot Runners (LoRR) `https://www.leagueofrobotrunners.org/` sponsored by Amazon Robotics

RESEARCH EXPERIENCE

IDM Lab, University of Southern California *Aug 2023 - Now*
Research Intern, mentored by Prof. Sven Koenig

LIRA Lab, University of Southern California *Aug 2024 - Now*
Research Intern, mentored by Prof. Erdem Bıyık

Multi-Robot Group, Carnegie Mellon University *Sept 2022 - Aug 2023*
Research Intern, mentored by Prof. Jiaoyang Li

Advanced Agent Robotics Technology Lab, Carnegie Mellon University *Oct 2021 - Aug 2023*
Research Intern, mentored by Prof. Katia Sycara

- Used model-based method (World Model) to train Shared Modular Policies across multiple agents with diverse morphologies. This refined policy can adapt to new morphologies and tasks.
- Developed a communication program to facilitate data transfer across PyTorch, Unity ML-Agent API, Unity Mujoco plugin, and the Mujoco physical engine backend

AWARDS

Lambda AI Research Grant VESSL AI Research Grant
2025 YC Startup School Grant
2025 ZJU National Academic Forum Outstanding Presentation Award
2025 IEEE IES SYPA Travel Award
2024 Fall USC GSG Professional Development Fund
2024 Futurewei High Performance Intern Award
2023 Best Paper Candidate Award in MRS (Multi-Robot System)
2023 USC Viterbi School of Engineering Fellowship
2023 CMU Algorithms With A Purpose (AWAP) AI Competition 2nd runner-up
2018 IBM China Outstanding Student Scholarship
2018 ACM-ICPC Xian Asia-East Continent Final Bronze Medal
2018 ACM-ICPC Asia Beijing Regional Contest Bronze Medal
2018 ACM-ICPC Asia Qingdao Regional Contest Silver Medal
2016-2020 ShanghaiTech University Merit Student / Scholarship of Academic Excellence

PAPER LIST

ITA-LaCAM: A Complete and Scalable TAPF Solver via Assignment-Aware Configuration-Space Search
Yimin Tang, Han Zhang, Shao-Hung Chan, Junsoo Kim, Erdem Biyik, Sven Koenig, Jingkai Chen (**in review**)

Cube-Based Automated Storage and Retrieval Systems: A Multi-Agent Path Finding Approach
Yi Zheng **Yimin Tang**, Sven Koenig, T. K. Satish Kumar (**AAAI 2026** Workshop on Multi-Agent Path Finding, **in review**)

Search-Aided Joint Agent-Environment Reinforcement Learning for Robust Lifelong MAPF with Rotations
He Jiang, Jingtian Yan, Yulun Zhang, **Yimin Tang**, Tanishq Harish Duhan, Rishi Veerapaneni, Guillaume Adrien Sartoretti, Jiaoyang Li (**in review**)

Emerging Trends in Multi-Agent Path Finding from the League of Robot Runners
Daniel Damir Harabor, Zhe Chen, Yue Zhang, **Yimin Tang**, Sven Koenig, Jingjin Yu (**in review**)

Judgelight: Trajectory-Level Post-Optimization for MAPF via Closed-Subwalk Collapsing
Yimin Tang, Sven Koenig, Erdem Bıyık (**WAFR 2026**)

RAILGUN: A Unified Convolutional Policy for MAPF Across Different Environments and Tasks
Yimin Tang*, Xiao Xiong*, Jingyi Xi, Jiaoyang Li, Erdem Bıyık, Sven Koenig (**IROS 2025 Oral**, ICRA 2025 Workshop on Foundation Models, SoCS 2025 Extend Abstract)[PDF]

Accelerating Focal Search in Multi-Agent Path Finding with Tighter Lower Bounds
Yimin Tang, Zhenghong Yu, Jiaoyang Li, Sven Koenig (**IROS 2025 Oral**, RSS 2025 Workshop on Multi-Agent)[PDF]

Enhancing Lifelong Multi-Agent Path Finding with Cache Mechanism
Yimin Tang*, Zhenghong Yu*, Yi Zheng, T. K. Satish Kumar, Jiaoyang Li, Sven Koenig (**AAMAS 2025** Workshop on Autonomous Robots and Multirobot Systems, **in review**)

Enhancing Long Context Performance in LLMs Through Inner Loop Query Mechanism
Yimin Tang, Yurong Xu, Ning Yan, Masood Mortazavi (**NeurIPS 2024** Workshop on Adaptive Foundation Models)

ITA-ECBS: A Bounded-Suboptimal Algorithm for Combined Target-Assignment and Path-Finding Problem
Yimin Tang, Sven Koenig, Jiaoyang Li (**SoCS 2024**)[PDF]

Solving Multi-Agent Task Assignment and Path Finding with a Single CBS Tree
Yimin Tang, Zhongqiang Ren, Jiaoyang Li, Katia Sycara (**IEEE MRS 2023**, **Best Paper Candidate**, also presented at ICAPS 2024)

Multi-modal perception and transfer learning for grasping transparent and specular objects
Thomas Weng, Amith Pallankize, **Yimin Tang**, Oliver Kroemer, David Held (**IEEE RA-L** presented at ICRA 2020) [WEB]